

Modelado de rendimientos financieros: volatilidad condicional, colas pesadas y dependencia multivariada

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Agenda

- 1 Objective and Data
- 2 Empirical Characteristics of Returns
- 3 Volatility Modeling
- 4 Shock Modeling
- 5 Forecasting and Evaluation
- 6 Copula Multivariate Model
- 7 Conclusion

- ▶ Provide a coherent, step-by-step **recipe** for financial return modeling at the undergraduate level.
- ▶ Follow one consistent path: empirical characterization, univariate modeling, forecast evaluation, and multivariate extension.
- ▶ Keep a dual lens throughout:
 - **Statistical validity** of model assumptions.
 - **Risk-management usefulness** (VaR/ES forecasts).

Data (Training set)

- ▶ Daily returns: S&P 500 and 10-year U.S. Treasury note index (SPBDU1BT).
- ▶ Sample used across chapters: Jan 2014–Dec 2024.

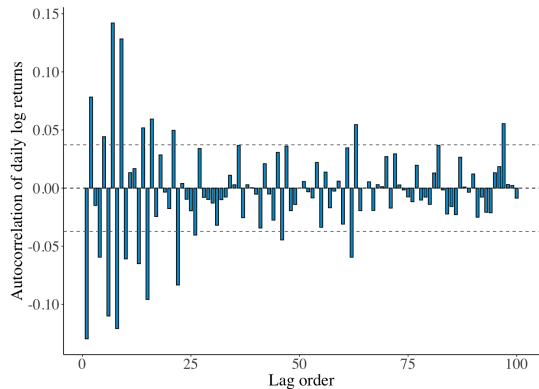
Forecast (Test set)

- ▶ Out-of-sample window: Jan 2025–Apr 2025 (81 observations).
- ▶ Forecasts: 1-day-ahead density, VaR and ES (2.5%).
- ▶ Tests: Christoffersen tests, PIT diagnostics, KS, and DM comparison.

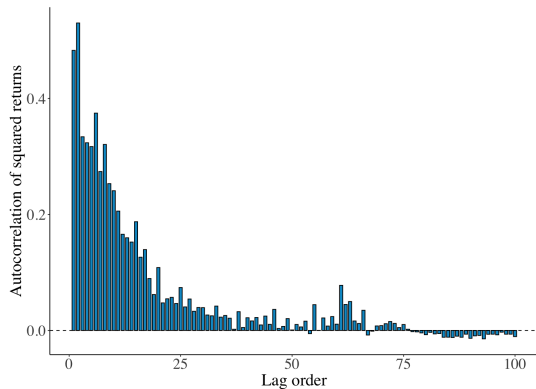
Empirical Characteristics of Returns

- ▶ Stylized facts on S&P 500 daily returns (2014–2024):
 - Near-zero autocorrelation in raw returns.
 - Volatility clustering in squared returns.
 - Leverage effect.
 - Heavy tails, asymmetry, and high kurtosis.
 - Time-varying cross-asset dependence.
- ▶ **These empirical characteristics serve as guidelines for what models should capture and replicate**

Autocorrelation Patterns



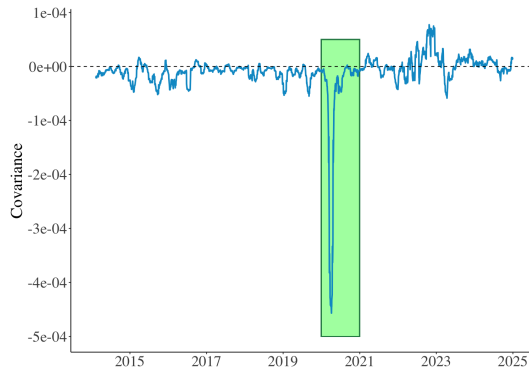
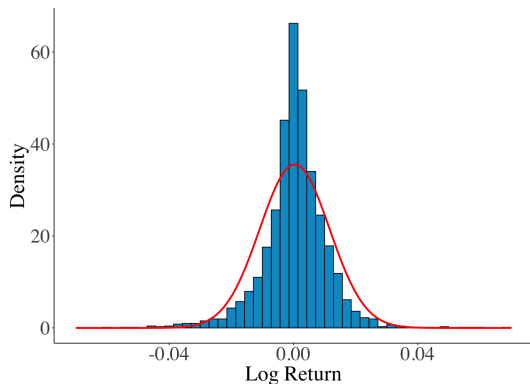
Returns ACF



Squared returns ACF

- ▶ Returns: little linear predictability.
- ▶ Squared returns: clear dependence and clustering.

Distributional and Dependence Features



- ▶ The empirical distribution exhibits heavier tails than the normal distribution and asymmetry.
- ▶ The correlation among assets appears to vary over time.

Return model

$$R_{t+1} = \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim \text{i.i.d. } D(0, 1)$$

- ▶ Goal: model conditional variance σ_{t+1}^2 .
- ▶ Ljung-Box on R_{t+1}^2 : strong evidence of conditional heteroskedasticity.
- ▶ Two models:

$$\mathbf{GARCH(1,1):} \quad \sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2$$

$$\mathbf{GJR-GARCH(1,1):} \quad \sigma_{t+1}^2 = \omega + \alpha R_t^2 + \alpha \theta I_t R_t^2 + \beta \sigma_t^2$$

- ▶ Indicator variable: $I_t = \mathbf{1}\{R_t < 0\}$.
 - If $R_t < 0$, $I_t = 1$ and shock impact is $\alpha(1 + \theta)R_t^2$.
 - If $R_t \geq 0$, $I_t = 0$ and shock impact is αR_t^2 .
- ▶ GJR term ($\theta > 0$) captures asymmetric (leverage) response.

Parameter Estimation: QMLE

- ▶ Parameters: $\vartheta = (\omega, \alpha, \beta)$ for GARCH(1,1), $\vartheta = (\omega, \alpha, \beta, \theta)$ for GJR-GARCH(1,1).
- ▶ **Quasi**-maximum likelihood: apply the normal MLE even if z_{t+1} is not normal (consistent estimates under correctly specified mean and variance).

Given a sample $\{R_t\}_{t=1}^T$, the conditional log-likelihood under normality is

$$\ell(\vartheta) = -\frac{1}{2} \sum_{t=1}^T \left[\log(2\pi) + \log(\sigma_t^2) + \frac{R_t^2}{\sigma_t^2} \right].$$

The QMLE estimator solves

$$\hat{\vartheta} = \arg \max_{\vartheta \in \Theta} \ell(\vartheta),$$

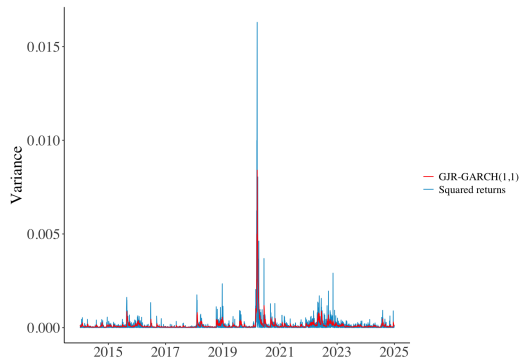
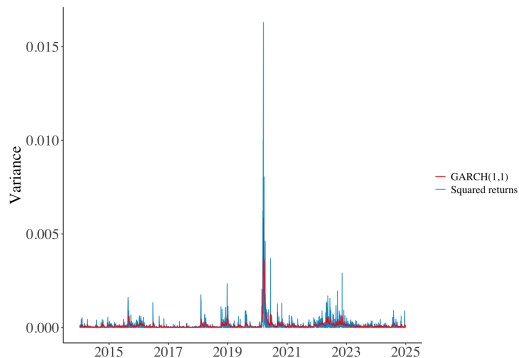
subject to positivity and finite long-run variance restrictions.

- ▶ No closed-form maximizer: solved numerically, each ϑ generating a full path $\{\sigma_t^2\}_{t=1}^T$ via the volatility recursion.

Estimated Dynamics (QMLE)

$$\text{GARCH}(1,1): \sigma_{t+1}^2 = 0.0000039 + 0.171R_t^2 + 0.794\sigma_t^2$$

$$\text{GJR-GARCH}(1,1): \sigma_{t+1}^2 = 0.0000040 + 0.054R_t^2 + 0.248I_tR_t^2 + 0.795\sigma_t^2$$

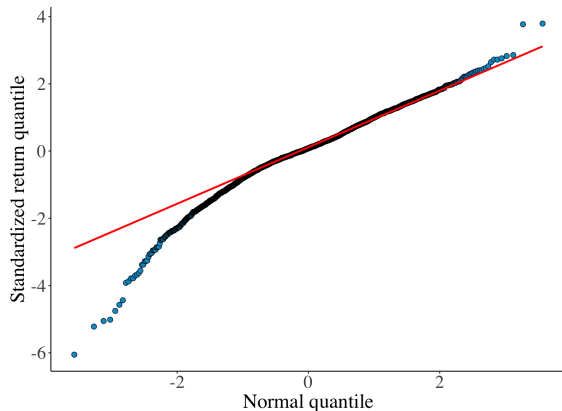
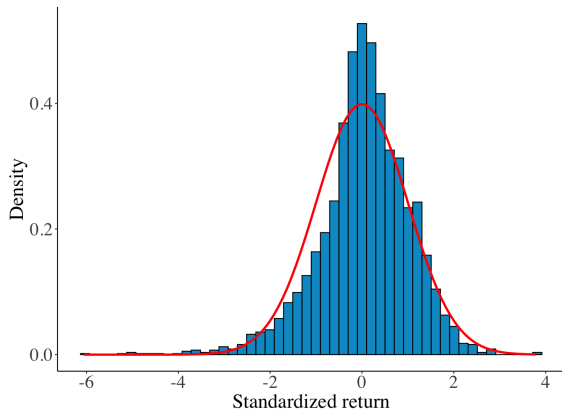


Key Takeaways on Volatility Modeling

Model	Mean QLIKE
GARCH(1,1)	1.6011
GJR-GARCH(1,1)	1.5673

- ▶ LR test for extra GJR parameter: $p < 1\%$ (Christoffersen (2012) framework).
- ▶ Both models absorb squared-return dependence.
- ▶ Persistence is close to one, consistent with financial volatility behavior.

Why Move Beyond Normal Shocks?



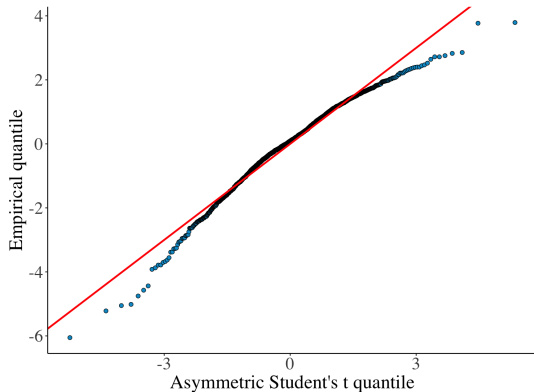
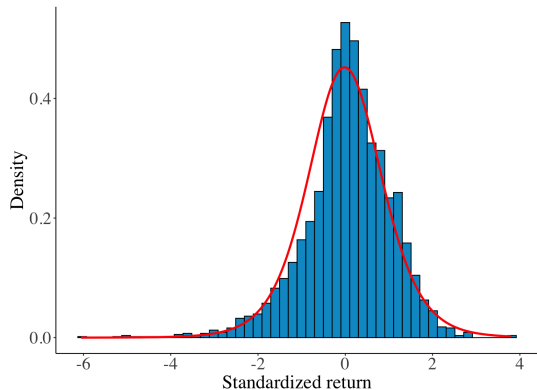
- ▶ Standardized residuals still show heavy tails and asymmetry.
- ▶ Normality is strongly rejected (very small p-values in diagnostics).
- ▶ Two alternatives considered: Asymmetric t and EVT-POT.

Approach 1: Asymmetric t Shocks

$$R_{t+1} = \sigma_{t+1} z_{t+1}, \quad z_{t+1} \sim \text{i.i.d. asyt}(d_1, d_2)$$

- ▶ Provides a skewed extension of the Student's t-distribution, enabling the modeling of both kurtosis and skewness observed in standardized returns.
- ▶ Parameters estimated after volatility QMLE step.
- ▶ Better center/tail fit than normal, but left tail remains underfit.

Asymmetric t Fit Diagnostics



- ▶ Good fit in the center and moderate tails.
- ▶ Left extreme tail remains the main weakness for risk management.

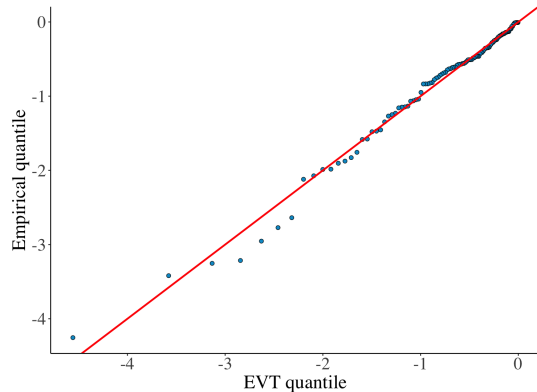
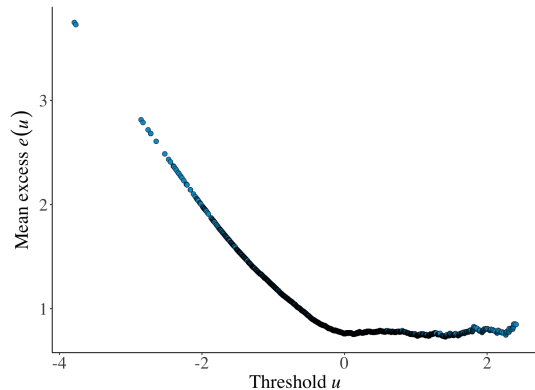
Approach 2: EVT (Peak-over-Threshold)

$$-z_{t+1} \mid (-z_{t+1} > u) \sim \text{GPD}(\xi, \beta, u), \quad u = 1.8$$

$$f(y) = \frac{1}{\beta} \left(1 + \frac{\xi(y-u)}{\beta} \right)^{-(1+\frac{1}{\xi})}, \quad y > u$$

- ▶ Focuses explicitly on left-tail risk, the critical region for VaR/ES.
- ▶ Threshold chosen through mean excess function linearity.

EVT Diagnostics



- ▶ Mean excess plot supports $u = 1.8$.
- ▶ GPD improves fit of the left tail versus full-distribution alternatives.

Estimated Return Models

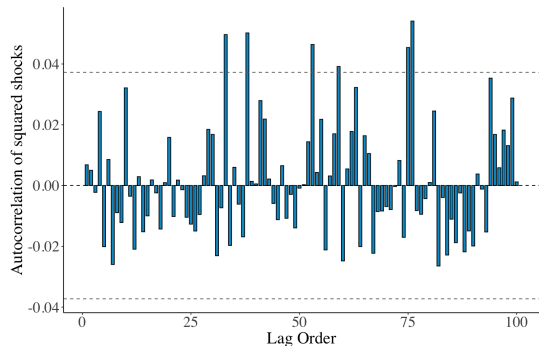
Model	α	β	θ	Tail params
GARCH-Asyt	0.171	0.794	–	$d_1 = 7.35, d_2 = 0.0115$
GJR-Asyt	0.054	0.795	4.62	$d_1 = 7.83, d_2 = 0.0129$
GARCH-EVT	0.171	0.794	–	$\beta_{\text{GPD}} = 0.774, \xi = 0.029$
GJR-EVT	0.054	0.795	4.62	$\beta_{\text{GPD}} = 0.621, \xi = 0.206$

- Four complete univariate return models are ready for forecasting.

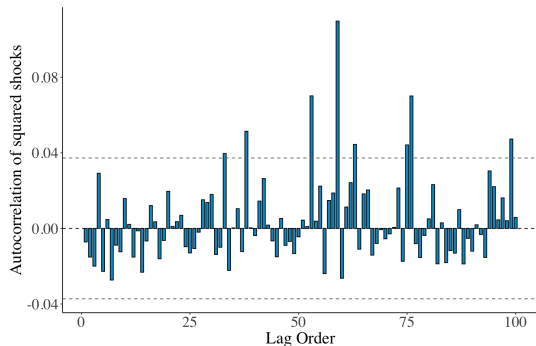
Forecasting and Evaluation Framework

- ▶ Forecast horizon: 1 day ahead, test window Jan–Apr 2025.
- ▶ Outputs: σ_{t+1}^2 , density forecast, $\text{VaR}_{2.5\%}$, $\text{ES}_{2.5\%}$.
- ▶ Evaluation blocks:
 - Volatility diagnostics (ACF of standardized squares, QLIKE, LR).
 - Risk diagnostics (Christoffersen LR_{uc} , LR_{ind} , LR_{cc}).
 - Density diagnostics (PIT histograms, KS, Berkowitz, DM).

ACF of Standardized Squares



GARCH(1,1)



GJR-GARCH(1,1)

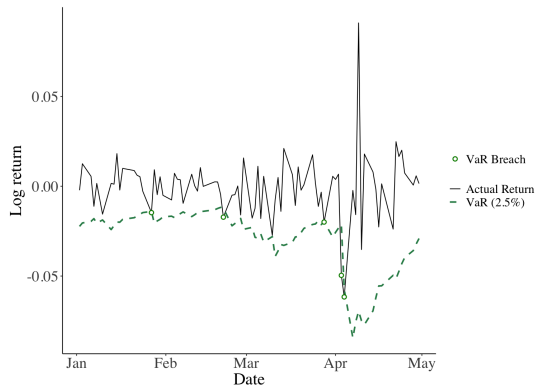
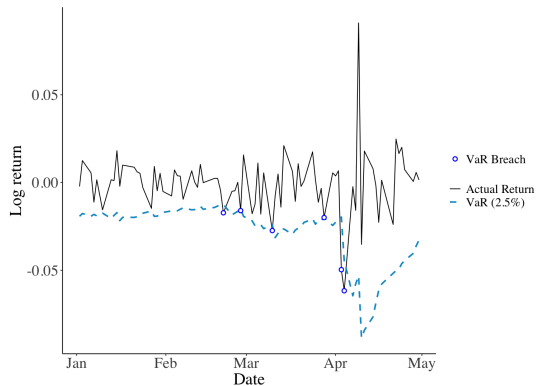
- ▶ Diagnostic goal: remove autocorrelation in standardized squared returns.
- ▶ Both models substantially absorb the dependence structure.

Asymmetric t Forecasts: VaR and ES

$$\text{VaR}_{t+1}^p = -\sigma_{t+1} F_{\text{asyt}}^{-1}(p; d_1, d_2), \quad \text{ES}_{t+1}^p = -\sigma_{t+1} \text{ES}_{\text{asyt}}(p)$$

- ▶ Full-density models provide direct VaR and ES forecasts.
- ▶ Calibration check at $p = 2.5\%$: expected violations about 2 in 81 days.

Asymmetric t VaR Breaches (Test Period)



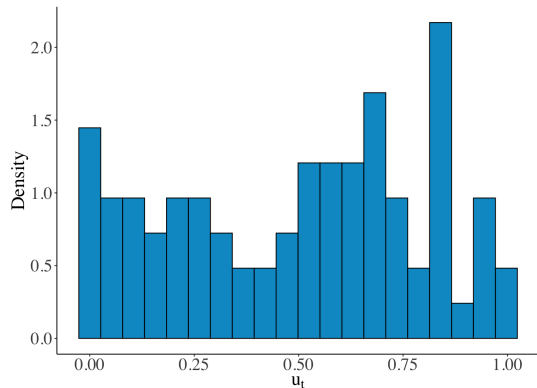
► Observed breaches exceed expectation and cluster after volatility shock.

Asymmetric t Backtesting (VaR)

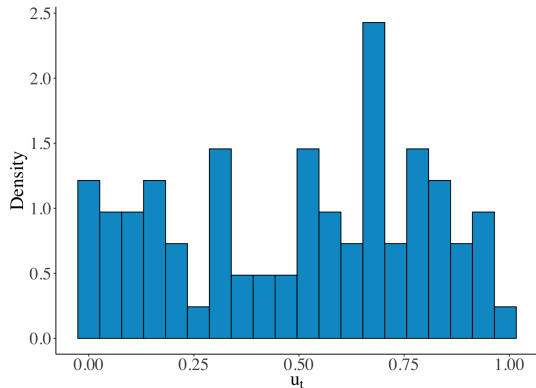
Test	p-value type	GARCH-Asyt	GJR-Asyt
LR_{uc}	Simulated	0.003	0.144
LR_{ind}	Simulated	0.095	0.047
LR_{cc}	Simulated	0.015	0.028

- ▶ VaR performance is weak: coverage and independence issues.
- ▶ Simulated LR_{uc} and LR_{cc} p-values are low for both models.
- ▶ Breaches cluster around volatility jumps, violating independence.

Asymmetric t Density Checks



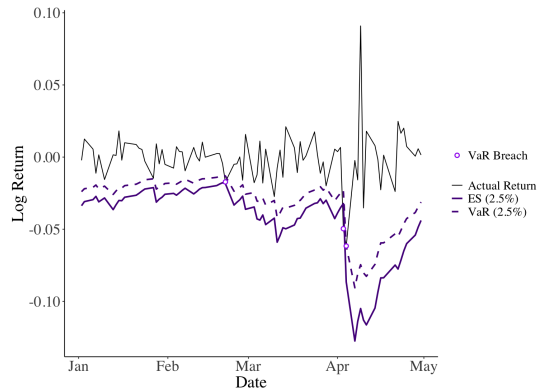
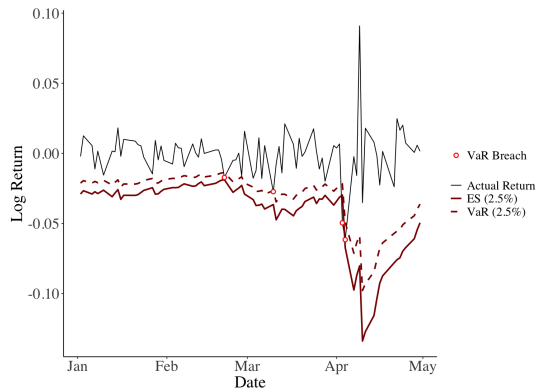
PIT histogram (GARCH-Asyt)



PIT histogram (GJR-Asyt)

- ▶ KS on $\Phi^{-1}(u_{t+1})$: 0.8512 (GARCH-Asyt), 0.7931 (GJR-Asyt).
- ▶ DM comparison: $p = 0.015$, favoring GJR-Asyt density forecasts.

EVT Forecasts in the Test Period



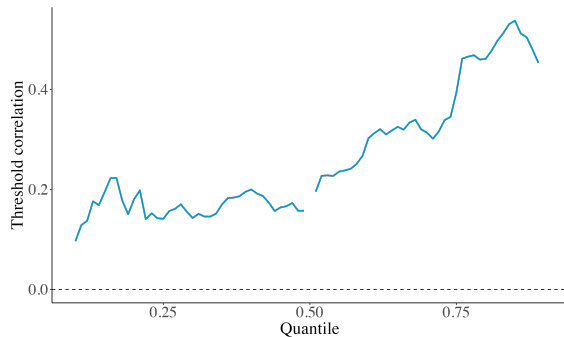
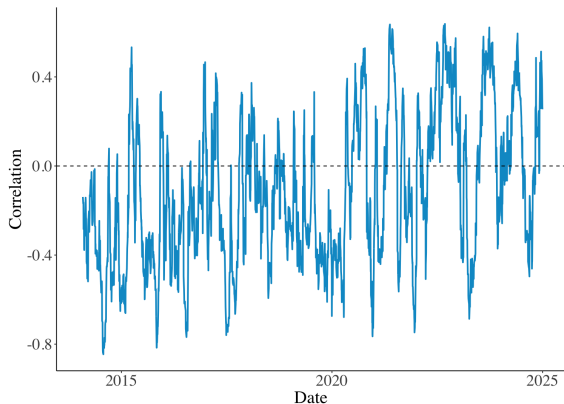
- ▶ EVT focuses directly on the left tail, giving more conservative risk forecasts.
- ▶ EVT variants show fewer breaches than Asyt counterparts.

EVT Backtesting Summary

Test	p-value type	GARCH-EVT	GJR-EVT
LR_{uc}	Simulated	0.188	0.554
LR_{ind}	Simulated	0.030	0.013
LR_{cc}	Simulated	0.217	0.200

- ▶ Coverage improves, but independence of violations remains problematic.
- ▶ Main weakness: slow adaptation to abrupt volatility regime changes.
- ▶ Stress-testing (2008 episode) supports stronger breach control for EVT variants.

Why Copulas?



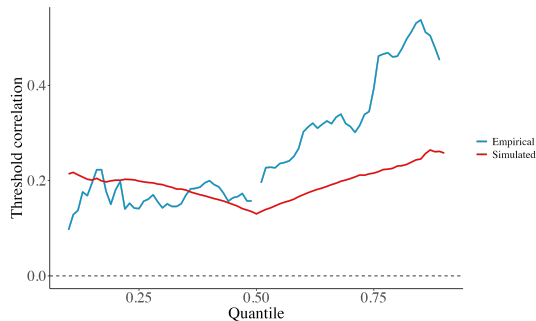
- ▶ Need dependence modeling for active risk management and diversification analysis.
- ▶ DCC captures dynamic linear dependence; copulas also target nonlinear tail dependence.
- ▶ Threshold-correlation evidence motivates copula-based modeling.

$$G(u_1, u_2; \rho^*, d) = t_{(d, \rho^*)}(t^{-1}(u_1; d), t^{-1}(u_2; d))$$

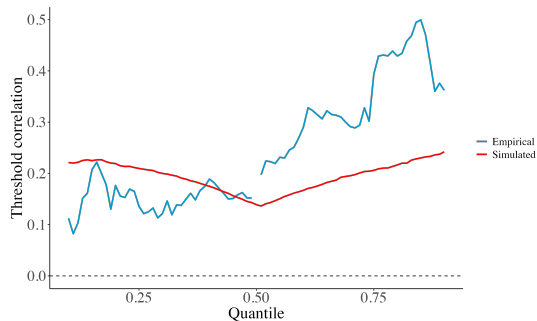
Copula model	ρ^*	d
GARCH-GARCH	-0.2121	4.9671
GJR-GJR	-0.2088	4.8304

- ▶ Static t-copula chosen for tractability and tail-thickness flexibility.
- ▶ Marginals estimated first (GARCH-Asyt / GJR-Asyt), then copula parameters.

Threshold-Correlation Fit



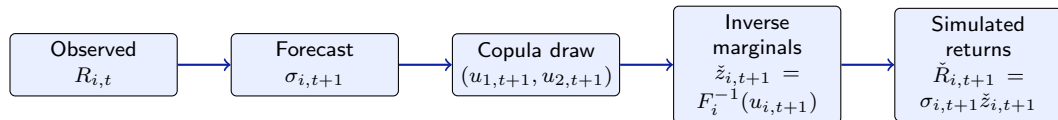
GARCH-GARCH copula



GJR-GJR copula

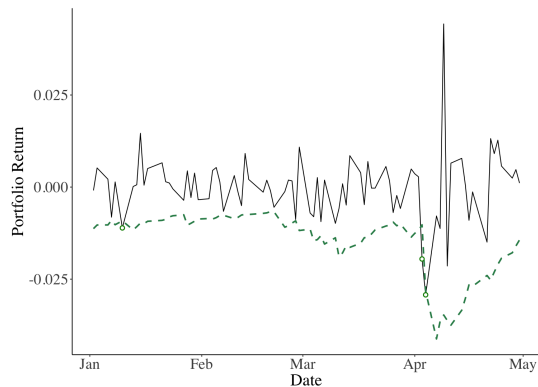
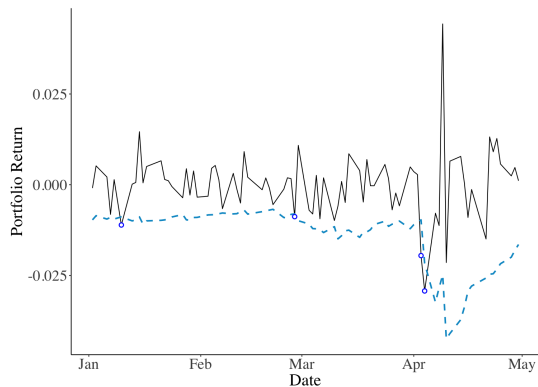
- ▶ t-copula reproduces heavier dependence in both tails reasonably well.
- ▶ Right-tail asymmetry in data is only partially captured (copula is symmetric).

Monte Carlo Forecasting Pipeline



- ▶ Repeat M times to form portfolio-return density.
- ▶ Compute VaR/ES from simulated quantiles and tail means.

Copula Portfolio Results



- Breaches are similar across the two copula variants.
- Main weakness remains clustering under abrupt volatility jumps.

Copula Backtesting and Density Validation

Test	p-value type	GARCH-GARCH	GJR-GJR
LR_{uc}	Simulated	0.188	0.554
LR_{ind}	Simulated	0.030	0.013
LR_{cc}	Simulated	0.217	0.200

- ▶ KS on normal transform: $p = 0.5271$ (GARCH-GARCH), $p = 0.4872$ (GJR-GJR).
- ▶ Density forecasts are statistically acceptable despite VaR clustering issues.

Main Conclusions

- ▶ The thesis delivers a coherent end-to-end framework from univariate to multivariate risk modeling.
- ▶ Volatility modeling: GJR improves in-sample volatility fit (QLIKE, LR evidence).
- ▶ Shock modeling:
 - Asymmetric t improves full-density fit.
 - EVT improves left-tail conservativeness for risk measures.
- ▶ Backtesting reveals a persistent challenge: **independence of violations** under sudden volatility spikes.
- ▶ Copula extension adds useful dependence structure and valid density forecasts for portfolios.

- ▶ Static dependence and symmetric copula limit adaptation and asymmetry capture.
- ▶ EVT tails improve coverage but still react slowly to abrupt regime shifts.
- ▶ Promising extensions:
 - Dynamic/asymmetric copulas (Christoffersen and Langlois (2011), Patton and Oh (2011)).
 - Shock models with time-varying skewness and kurtosis.
 - Broader stress-testing and alternative volatility proxies.

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Questions

 github.com/ferpmillan/thesis

 github.com/ferpmillan/thesis-tex